

TEAM HASHIRA · SIH26174 · ISRO

Experiment catalogue & capability guide

Twenty-one experiments SARTHI can supervise — fourteen for a space station, seven on Earth — with the props, every step, what the system checks, and the exact builder settings. Plus what SARTHI can and cannot see, how to run a demo, and how to teach it new objects.

21

experiments

4

built in, ready to run

13

body actions, left /
right

0

network calls — fully
offline

1 · What SARTHI can check

An experiment is a list of steps. Each step asks for evidence the camera can see — an object, what a hand does with it, a body action — and SARTHI judges the steps in order, speaks the next one, alerts on mistakes and writes a timestamped, tamper-evident log.

CAPABILITY	WHAT IT MEANS	STATUS
See objects	80 everyday objects out of the box (bottle, cup, bowl, phone, laptop, backpack, fruit, plant, scissors...) plus any object you train	● Reliable
Several objects at once	Every object the experiment names, in the same frame	● Reliable
“Show it”	The object held up close to the camera (fills 6 % of the frame or more)	● Reliable
“Hold it” — which hand	A hand on the object; a step can require the left or the right hand	● Reliable
“Move it”	The object carried a quarter of the way across the picture while held	● Reliable
“Put it in its place”	The object standing in a marked spot (the saffron box) for 2 s — procedure files only	● Reliable
“Pour it”	The object held and tipped over. Needs the object trained with pouring photos: the stock detector loses a tipped bottle	● With care
Body postures	Raise a hand, hand to face, hand on head, reach out, both hands up, hands together, arms crossed, arms out, hands on hips (standing)	● Reliable
Movements	Wave, lift, lower, clap — read from the last 2–2.5 s; do them slowly (a laptop reads 3–4 frames a second)	● With care
Movement while holding	Hold it + a movement, e.g. lift the bottle with the right hand	● Reliable
Any orientation	Postures are measured against the body's own axis — upright, lying or upside down read the same	● Reliable
Alerts (spoken)	Wrong object, wrong hand, stall, cannot verify — always. Skip and out of sequence — in Strict mode	● Reliable
Record	Timestamped, tamper-evident step log; local video; RTSP stream to an IP	● Reliable

HONEST LIMITS

- Fingers: thumbs-up, pinch, counting fingers — the pose model has wrists, not fingers.
- Fine manipulation: twisting a cap one way or the other, cutting, writing.
- Whether liquid actually flowed — only that the container was tipped while held.
- Small or flat objects are unreliable with the stock model: books, spoons, pens, pills.
- Fast movements on a laptop: a quick wave or clap falls between frames.
- Two objects in one builder step (a procedure file can do it).
- Telling two similar objects apart (two medicine bottles) — train each one.

2 · Body actions

Read from the skeleton (17 body points) and measured against the body's own axis — shoulders to head, hips to shoulders — never the image, so a crew member upside down raises a hand exactly as one standing does. One-hand actions can be required for the **left** or the **right** hand; the other hand raises a spoken **wrong hand** alert.

ACTION	KIND	HOW TO DO IT FOR THE CAMERA
Raise a hand	One hand	Forearm pointing up
Hand to face	One hand	Wrist at the mouth or nose — drinking, eating
Hand on head	One hand	Elbow up, hand beside the head
Reach out	One hand	Arm straight, lifted away from the body
Wave	One hand · movement	Raised hand going side to side, at least twice
Lift	One hand · movement	Hand rising half a shoulder-width or more
Lower	One hand · movement	Hand coming down the same distance
Raise both hands	Both	Both forearms up
Hands together	Both	Wrists close together
Cross your arms	Both	Each hand past the middle of the chest
Arms out	Both	A T: arms straight at shoulder height (stand back from the camera)
Hands on hips	Both	Standing, knees in view
Clap	Both · movement	Hands closing again and again — slowly

DOING IT WELL

- Sit 60–80 cm from the laptop; chest, **both elbows** and the table in view.
- Hold each action about two seconds; do one thing at a time and wait for the voice.
- Movements slowly and widely: a laptop reads 3–4 frames a second.
- Hold objects so they fill a good part of the frame for “show it”.

ALERTS IT SPEAKS

- **Wrong object** — picking up what a later step needs.
- **Wrong hand** — the other hand doing a left/right step.
- **Out of sequence / skipped** — in Strict mode.
- **No progress** — a step left undone for 60–120 s.
- **Cannot verify** — evidence too weak to trust.

3 · Running and building an experiment

START In the Clippy folder: `uv run python scripts/web_demo.py`, then open `http://localhost:8000`.

RUN A BUILT-IN ONE

1. **Procedures** tab → *Built-in* → the experiment's card → **Load**.
2. **Mission** tab → choose **Clean** (normal) or **Strict** (also flags steps done early) → **New run**.
3. Follow the voice. An alert covers the top of the video: **Acknowledge** (or Esc).
4. At the end a debrief appears; **End run** releases the camera.
5. **Archive** tab → the run → steps, alerts, video, **Verify chain**.

BUILD YOUR OWN

1. **Procedures** → *Build an experiment* → type a name.
2. Add steps from **Objects** (80 stock + your trained ones, listed first) or **Body actions**.
3. In each step choose what is done with the object — **Show / Hold / Pour / Move** — and **which hand** — *Either / Left / Right*.
4. Leave the wording empty to use the default shown in grey, or type your own.
5. **Save** (it appears under *Your experiments*, even after a restart) or **Save & run**.

“Put it back in its place” and “two objects in one step” need a procedure file — copy `procedures/seed_germination.yaml` and edit it.

4 · Teaching SARTHI a new object

1. **Models** tab → type the object's name → **+**. Keep the **background** class.
2. Capture from the live laptop camera in the room you will run in: **40–60 photos** per object or state (5–6 bursts of ×10, moving it between bursts) — different angles, distances, both hands, standing and held. **Background: 60–80** — the empty table, empty hands, a finger, your face, and every *other* prop on its own.
3. **Train** → **Objects** → **Find the object in my photos** → check the boxes in **Boxes** (click a wrong one to leave it out) → 15 epochs → **Train object detector** (≈15–20 min on a laptop).
4. **Test model** live → **Deploy**. Your object joins the 80 stock ones and stays after a restart; deleting its class hides it again.

PHOTOS OR VIDEO? A video upload is cut into 40 still frames, so it is a faster way to collect varied photos — record one slow 15–20 s clip per class. Its held-out accuracy looks better than it is (neighbouring frames are near-copies); judge it with **Test model**. For a pouring step, include photos of the object tipped at 30–90°.

5 · Space station experiments

01 Seed germination check

Plant biology **Built-in — ready to run** **Reliable** procedures/seed_germination.yaml

ISRO's Axiom-4 experiment sprouted moong and methi seeds on the ISS. Growing food is how a BAS crew will one day feed itself; the daily check is simple and easy to get wrong.

PROPS **Bowl of sprouted moong (or a small potted plant)** (seed tray) · **Water bottle** (water dispenser) · **Phone** (crew tablet)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Take the tray out of the growth chamber	A hand on the tray	bowl / potted plant · Hold it
2	Water the seeds	Bottle in hand, tray in view	bottle · Hold it (+ tray visible)
3	Photograph the growth	Phone in hand	cell phone · Hold it
4	Return the tray to the growth chamber	Tray standing in the saffron GROWTH_CHAMBER box for 2 s	place – procedure file
5	Report complete	Both hands raised	Raise both hands

ALERTS TO SHOW Wrong object (pick up the phone at step 2) · Out of order in Strict (return the tray before the photo) · Stall

NOTES Check in the camera preview that the bowl/plant gets a box. The recommended main demo.

02 Potable water sampling

Environmental health **Built-in — ready to run** **Reliable** procedures/water_sample.yaml

Station crews sample the drinking water for microbes and log the result; a wrong container or a skipped log entry matters when the ground is not watching.

PROPS **Water bottle** (sample container) · **Cup or mug** (sampling cup) · **Phone** (crew tablet)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Report ready	A raised hand	Raise a hand
2	Retrieve the sample container	Bottle held up to the camera	bottle · Show it
3	Collect the sample	Bottle AND cup held up together	two objects in one step – procedure file
4	Log the sample on the tablet	Phone held up	cell phone · Show it
5	Confirm completion	Both hands raised	Raise both hands

ALERTS TO SHOW Wrong object (show the tablet at step 2) · Out of order in Strict (log before collecting)

NOTES Shows two objects at once. Good backup demo.

03 Crew fitness and hydration check

Human physiology

Built-in — ready to run

Works with care

procedures/crew_fitness.yaml

In microgravity muscle and bone waste away, so crews exercise every day and drink on a schedule. The most human-activity demo: every step is read from the body.

PROPS **Water bottle** (crew water ration)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Raise your RIGHT hand to report ready	Right hand raised	Raise a hand · Right hand
2	Cross your arms to stretch	Arms crossed	Cross your arms
3	Wave your LEFT hand, slowly	Left hand waving	Wave · Left hand
4	Lift the bottle with your RIGHT hand	Right hand on the bottle AND lifting	bottle · Hold it + Lift · Right hand
5	Drink the water	Hand at the face, bottle in view	bottle + Hand to face
6	Raise both hands to finish	Both hands raised	Raise both hands

ALERTS TO SHOW Wrong hand (raise the LEFT hand at step 1) · Stall

NOTES Keep both elbows in frame; wave slowly and widely (a laptop reads 3–4 frames a second).

04 Crew meal and hydration log

Nutrition

Build it in the builder (≈2 min)

Works with care

Keeping crew intake on schedule is a daily medical requirement; Gaganyaan's food system is ISRO's own.

PROPS **Banana or apple** (meal pack) · **Water bottle** (drink) · **Phone** (crew tablet)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Take the meal pack	Hand on the fruit	banana · Hold it
2	Eat	Hand at the face holding the fruit	banana · Hold it + Hand to face
3	Drink	Hand at the face holding the bottle	bottle · Hold it + Hand to face
4	Log the intake	Phone held up	cell phone · Show it
5	Put the leftover away	Hand lowering the fruit	banana · Hold it + Lower

ALERTS TO SHOW Wrong object (bottle before the meal) · Out of order in Strict

NOTES The same gesture means two steps depending on the object in hand.

05 Biological sample to cold storage

Life sciences

Needs a procedure file (has a “put back in its place” step)

Reliable

Blood, saliva and cell samples must reach the freezer quickly and in the right order, or the science is lost.

PROPS **Water bottle** (sample tube) · **Cup** (transfer holder) · **Phone** (crew tablet) · **A marked spot on the table** (freezer)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Report ready	A raised hand	Raise a hand
2	Take the sample tube	Hand on the bottle	bottle · Hold it
3	Place it in the transfer holder	Hand lowering the bottle (or bottle + cup together)	bottle · Hold it + Lower
4	Put it in the freezer	Bottle in the freezer spot for 2 s	place – procedure file
5	Log it	Phone held up	cell phone · Show it

ALERTS TO SHOW Out of order in Strict (logging before storing) · Wrong object

NOTES Needs a small procedure file for the freezer spot.

06 Fluid transfer in microgravity

Physical sciences

Build it in the builder (≈2 min)

Needs training for “pour”

Moving liquids in microgravity is a real payload task; the crew must pour, record and stow in order.

PROPS **Water bottle** (source container) · **Cup** (measuring beaker) · **Phone** (crew tablet)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Pick up the source container	Hand on the bottle	bottle · Hold it
2	Pour into the beaker	Bottle in hand AND tipped over	bottle · Pour it
3	Record the volume	Phone held up	cell phone · Show it
4	Stow the beaker	Cup carried across the picture	cup · Move it

ALERTS TO SHOW Wrong object · Out of order in Strict

NOTES The stock detector loses a tipped bottle (1 of 10 at 45°). Train your bottle with pouring photos, or use “bottle and cup together”.

07 Emergency response drill

Crew safety

Build it in the builder (≈2 min)

Reliable

ISS crews rehearse alarm drills regularly; on BAS nobody on the ground talks them through it.

PROPS **Backpack** (emergency kit) · **Phone** (comms tablet)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Acknowledge the alarm	Right hand raised	Raise a hand · Right hand
2	Grab the emergency kit	Hand on the backpack	backpack · Hold it
3	Carry it to the hatch	Backpack carried across the picture	backpack · Move it
4	Report on the comms tablet	Phone held up	cell phone · Show it
5	Crew accounted for	Both hands raised	Raise both hands

ALERTS TO SHOW Wrong hand · Wrong object · Stall (a frozen crew member)

NOTES Backpacks and phones are detected very well.

08 Tool accountability

Payload maintenance

Build it in the builder (≈2 min)

Works with care

In space a loose tool floats away and becomes a hazard: every tool taken out must come back.

PROPS **Scissors** (the tool) · **Phone** (crew tablet)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Take the tool from the kit	Hand on the scissors	scissors · Hold it
2	Do the repair	Both hands together, tool in hand	scissors · Hold it + Hands together
3	Return the tool	Hand lowering the scissors	scissors · Hold it + Lower
4	Confirm on the tablet	Phone held up	cell phone · Show it

ALERTS TO SHOW Wrong object · Out of order in Strict

NOTES Scissors are only moderately well detected — test in the preview first.

09 Science data downlink

Payload operations

Build it in the builder (≈2 min)

Reliable

Experiment data goes to the ground in a scheduled window; a missed step means lost science.

PROPS **Laptop** (payload computer) · **Phone** (data drive)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Open the payload computer	Hand on the laptop	laptop · Hold it
2	Connect the data drive	Hand on the phone	cell phone · Hold it
3	Start the transfer	A raised hand	Raise a hand
4	Stow the drive	Phone carried across the picture	cell phone · Move it
5	Downlink complete	Both hands raised	Raise both hands

ALERTS TO SHOW Wrong object · Wrong hand

10 Cabin stowage audit

Logistics

Build it in the builder (≈2 min)

Reliable

Before undocking everything must be stowed and counted; loose items are dangerous.

PROPS **Bottle and cup** (loose items) · **Backpack** (stowage bag) · **Phone** (crew tablet)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Stow the bottle	Bottle carried across	bottle · Move it
2	Stow the cup	Cup carried across	cup · Move it
3	Close the bag	Hand on the backpack	backpack · Hold it
4	Report	Phone held up	cell phone · Show it

ALERTS TO SHOW Wrong object (packing in the wrong order) — the whole point

11 Food sample decay study

Food science

Needs a procedure file (has a "put back in its place" step)

Reliable

How long food stays safe matters for long missions such as BAS and the Moon.

PROPS **Banana or apple** (food sample) · **Bowl** (sample dish) · **Phone** (crew tablet) · **A marked spot** (incubator)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Take the sample out	Hand on the fruit	apple · Hold it
2	Put it in the sample dish	Hand lowering the fruit	apple · Hold it + Lower
3	Photograph it	Phone in hand	cell phone · Hold it
4	Return it to the incubator	Fruit in the incubator spot for 2 s	place – procedure file

ALERTS TO SHOW Out of order in Strict · Stall

NOTES A close cousin of the germination check.

12 Waste management

Life support

Build it in the builder (≈2 min)

Reliable

On a station every piece of trash must be bagged, or it floats into the air system.

PROPS **Bottle and cup** (waste) · **Backpack** (waste bag) · **Phone** (crew tablet)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Pick up the empty bottle	Hand on the bottle	bottle · Hold it
2	Bag it	Bottle carried across	bottle · Move it
3	Bag the used cup	Cup carried across	cup · Move it
4	Seal the bag	Both hands together on the bag	backpack · Hold it + Hands together
5	Log it	Phone held up	cell phone · Show it

ALERTS TO SHOW Wrong object · Out of order in Strict

13 The PS sample experiment — nested sample retrieval

SIH26174 brief Needs props + training Needs props + training procedures/proc_a.yaml

The problem statement's own example: a box containing a red and a yellow box. This is what the judges' brief describes.

PROPS **Shoebox** (outer container) · **Red and yellow small boxes** (sample boxes) · **Vial + tweezers + clip** (sample and tools) · **ArUco rack board** (payload rack)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Open the outer container	outer_box_open seen	procedure file
2	Remove the red box, place it on marker R	Hand on the red box, then near marker R	procedure file
3	Remove the yellow box, place it on marker Y	Hand on the yellow box, then near marker Y	procedure file
4	Open the red box	red_box_open seen	procedure file
5	Transfer the vial with tweezers	Tweezers on the vial, vial near the yellow box	procedure file
6	Close the red box and secure it	red_box_closed, secured	procedure file

ALERTS TO SHOW Skip · Out of order · Stall · Free-float (a tethered item adrift)

NOTES The procedure is built and tested; it needs the prop kit and a model trained on the boxes' open/closed states. The finale plan.

14 Drink water (your original experiment)

Human activity

Built-in — ready to run

Works with care

procedures/drink_water_body.yaml · drink_water.yaml

Pick up the bottle → open the cap → drink → close the cap → put it back. Ships in two forms.

PROPS Water bottle (the bottle)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Pick up the bottle	Hand on the bottle	bottle · Hold it
2	Open the cap	Body version: both hands together · Trained version: bottle_open seen	Hands together (body) / trained class
3	Drink	Hand at the face, bottle in view	bottle + Hand to face
4	Close the cap	Body version: hands together · Trained: bottle_closed seen	Hands together (body) / trained class
5	Put it back	Bottle in the HOME box for 2 s	place – procedure file

ALERTS TO SHOW Out of order in Strict (put back before closing) · Stall

NOTES Telling open from closed needs the cap state trained — hard at webcam distance. The body version needs no training.

6 · On Earth — the same engine, a different file

SARTHI is not specific to one experiment: a new procedure is a new file, with no code change. These show where the same system earns its keep on the ground.

15 Medicine routine for elderly people

Healthcare Build it in the builder (≈2 min) Reliable

Missed or double doses are a major cause of hospital visits, and family cannot watch all day.

PROPS **Bottle** (medicine bottle) · **Cup** (glass of water) · **Phone** (confirmation)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Take the medicine bottle	Hand on the bottle	bottle · Hold it
2	Take the pill	Hand at the face	Hand to face
3	Drink water	Hand at the face holding the cup	cup · Hold it + Hand to face
4	Put the bottle back	Hand lowering the bottle	bottle · Hold it + Lower
5	Confirm on the phone	Phone held up	cell phone · Show it

ALERTS TO SHOW Out of order in Strict · Stall (a missed dose)

NOTES Telling one medicine bottle from another needs each one trained.

16 Physiotherapy exercises at home

Rehabilitation Build it in the builder (≈2 min) Reliable

Patients skip or reorder prescribed exercises and the doctor never knows. No props at all.

PROPS **None** (just the body)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Raise your right arm	Right hand raised	Raise a hand · Right hand
2	Raise your left arm	Left hand raised	Raise a hand · Left hand
3	Cross your arms	Arms crossed	Cross your arms
4	Reach forward with your right arm	Right arm straight out	Reach out · Right hand
5	Wave to finish	A hand waving	Wave

ALERTS TO SHOW Wrong hand (the core of it) · Stall

NOTES Pure human activity recognition — the most reliable demo.

17 School chemistry lab procedure

Education & safety

Build it in the builder (≈2 min)

Needs training for "pour"

Students mix steps up, and in a lab the order can be a safety issue.

PROPS **Bottle** (reagent bottle) · **Cup** (beaker) · **Phone** (lab notebook)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Take the reagent	Hand on the bottle	bottle · Hold it
2	Pour into the beaker	Bottle in hand and tipped	bottle · Pour it
3	Record the reading	Phone held up	cell phone · Show it
4	Return the reagent	Hand lowering the bottle	bottle · Hold it + Lower

ALERTS TO SHOW Out of order in Strict (recording before pouring) · Wrong object

NOTES Train the reagent bottle with pouring photos for a reliable pour.

18 Airport security screening

Travel

Build it in the builder (≈2 min)

Reliable

The procedure everyone knows and everyone gets wrong — instantly relatable.

PROPS **Backpack** (cabin bag) · **Laptop** (laptop) · **Phone** (phone)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Put the bag down	Hand lowering the backpack	backpack · Hold it + Lower
2	Take out the laptop	Hand on the laptop, lifting	laptop · Hold it + Lift
3	Put the phone in the tray	Phone carried across	cell phone · Move it
4	Hands up for the scan	Both hands raised	Raise both hands

ALERTS TO SHOW Wrong object (phone before laptop) · Out of order in Strict

NOTES Every prop is detected very well.

19 Warehouse order packing

Logistics / e-commerce

Build it in the builder (≈2 min)

Reliable

Wrong items in a parcel cost e-commerce companies heavily.

PROPS **Bottle, cup, phone** (order items) · **Backpack** (parcel)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Pack item 1	Bottle carried across	bottle · Move it
2	Pack item 2	Cup carried across	cup · Move it
3	Pack item 3	Phone carried across	cell phone · Move it
4	Seal the parcel	Both hands together on the bag	backpack · Hold it + Hands together

ALERTS TO SHOW Wrong object — the whole point

20 Baby feed preparation

Childcare & hygiene

Build it in the builder (≈2 min)

Works with care

Hygiene order matters, and a tired parent at 3 a.m. skips steps.

PROPS **Bottle** (feeding bottle) · **Cup** (water) · **Bowl** (steriliser)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Take the bottle from the steriliser	Hand on the bottle	bottle · Hold it
2	Pour in the water	Cup in hand and tipped	cup · Pour it
3	Mix	Both hands together on the bottle	bottle · Hold it + Hands together
4	Test on the wrist / taste	Bottle at the face	bottle · Hold it + Hand to face
5	Put the steriliser away	Bowl carried across	bowl · Move it

ALERTS TO SHOW Out of order in Strict · Wrong object

NOTES "Pour" is more reliable with the cup trained on pouring photos.

21 Factory assembly checklist

Manufacturing

Build it in the builder (≈2 min)

Works with care

Factories and pharma plants must prove every step of a standard procedure was done, in order — the biggest real-world market for SARTHI.

PROPS Scissors (tool) · Cup (part) · Phone (QC scanner)

#	OPERATOR DOES	SARTHI CHECKS	BUILDER SETTINGS
1	Take the tool	Hand on the scissors	scissors · Hold it
2	Fit the part	Both hands together on the part	cup · Hold it + Hands together
3	Return the tool	Hand lowering the scissors	scissors · Hold it + Lower
4	Scan for quality check	Phone held up	cell phone · Show it

ALERTS TO SHOW Wrong object · Out of order in Strict — the tamper-evident log is the compliance proof

NOTES Scissors are only moderately well detected.

7 · Demo plan (internal hackathon)

1. **The story (30 s).** On BAS, ground support cannot watch every experiment live. SARTHI runs on board, offline, and supervises the crew step by step.
2. **Main demo — Seed germination check (Clean).** The full five steps with voice prompts; show the debrief.
3. **A mistake — wrong object.** New run; at step 2 pick up the phone: “Wrong object: the crew tablet is for step 3. Now: Water the seeds.”
4. **Human activity — Crew fitness check.** Raise the LEFT hand at step 1: “Wrong hand: use your right hand.” Then wave, lift the bottle with the right hand, drink.
5. **Out of sequence — Strict mode.** Germination again: return the tray before photographing it.
6. **The record.** Archive → verify the chain: a few kilobytes of verified log go to the ground, not a video stream.
7. **Close.** “A new experiment is a new file” — show this catalogue: space, healthcare, factories.

BEFORE THE DEMO Rehearse twice in the same room and light. Check in the preview that every prop gets a box. Keep one good run's video on disk. Backup demo: Potable water sampling.